

FURNACE Protocol — Complete

Specification · v5.3.1

The single source of truth is `furnace.config.ts` v5.3.1. This document explains it.

Every number below is a field in that file. A rebrand edits the config; this document then describes the new brand automatically.

Config hash:

`0x9ecedb63d5d10951caaf506032a1cf6d14b9e2196f236e2d2a45c200870463f7`
(recompute with `pnpm config:hash`; record on-chain at deploy)

0. The idea

FURNACE is an omnichain loyalty-staking protocol. Users stake liquidity, earn emissions weighted by how long they have stayed, and 80% of every fee the protocol touches burns the token or deepens its liquidity. The other 20% is named and fixed: 10% to ASH stakers in the Hearth, 10% to the treasury. No fee is ever paid to a team wallet. Nothing is ever locked: every timer starts when the user clicks, and the only instant, free, timer-free action is compounding. Time in the pool is the yield — and time is an asset: a position is a Deed, an NFT that can be sold without a single token leaving the pool. Five ideas carry the design. Everything else is a parameter.

1. Time is the yield. Heat grows on the square-root curve from 1.0× to 2.5× over a year, plus prorated milestone bumps to 3.0×. Early loyalty pays hardest; diminishing returns keep a two-year position from lapping a one-year one.

2. Fees never pay the team. Every fee is routed to burning EMBER or deepening permanent liquidity. The Hearth's 10% cut goes to ASH stakers as real yield, not the team.
3. Ashfall closes the loop. Prior-epoch burns re-emit to Flame+ stakers weighted by stake $\times$ Heat², at source-specific recycle rates. Sellers and leavers fund stayers.
4. Compound is the only door with no lock. Stoking is instant, free, and the only action that grows a hot position without resetting age — rewards compound back into the lot that earned them, at that lot's age.
5. Genesis creates liquidity, not sellers. Kindling is a clearing-price auction: 90% of the LP becomes depositors' auto-staked Deeds, 10% burns forever, atomic in the close transaction. Nobody gets loose tokens; the pool exists before the next block.

1. Tokens

Three things that look like tokens. Two of them are tokens.

EMBER — the protocol token

Property	Value
Hard cap	21,000,000
Standard	LayerZero OFT v2 (one canonical supply, all chains)
Decimals	18 (EVM) / 9 (Solana), 6 shared decimals for messaging
Home chain	Ethereum
OFT rate limit	250,000 / hour
Transfer tax	0% — forever. Fees live on protocol actions, never on transfers.

Emissions follow a Bitcoin-style halving: 21,600 EMBER/day at open (0.25/second network-wide), 365-day half-life, 1,000/day floor.

The 21M cap is a live-supply ceiling, not a cumulative-mint ceiling:

`mint(epoch) = min(targetEmission, 21M - totalSupply)`. The decay

curve hits

the 1,000/day floor around day 1,618 (~10.85M emitted); without burns, the

floor would mint past 21M roughly 29.7 years after launch. So the floor is a target, not a promise: burns reopen mint headroom, and the tail continues only while the deflation engine creates room. Late-stage emissions are structurally dependent on burns — the protocol must earn its tail. When headroom runs short, all recipients scale down pro-rata; the build rejects any config where `supplyCeiling.hardCap ≠ token.maxSupply`.

Allocations: 420,000 EMBER (2%) minted once for the Kindling seed; 500,000 to the treasury (no cliff, linear over 365 days); team allocation 0.

ASH — the share / governance token

Property	Value
Hard cap	70,000 — nothing else in the protocol can mint ASH
Launch share	41,000 Ignition rewards + 1,000 Kindling bonus
Forge share	28,000, trickled through staking over 730 days

The validator recursively sums every leaf of the ASH allocation tree and requires exactly 70,000 — the 71k overallocation bug (42k + 28k + a separate

1k Kindling pool) can never recur. Ignition's reward total and Kindling's bonus pool are each cross-checked against their tree leaf.

ASH staked in the Hearth earns a cut of every protocol fee plus governance weight. It is the productive share token: scarce, and it pays.

Heat — not a token

Heat is a per-position loyalty score: the stake-weighted age of your liquidity, converted to a yield multiplier.

```
age      = stored lot age, capped at 365 days (it never
counts past the cap)
lotHeat = 1.0 + 1.5 × √(age / 365) + bump(age)
bump     = milestone bumps, prorated linearly between
milestones (max +0.5×)
Heat(position) = Σ(lot LP × lotHeat) / Σ(lot LP)
```

The curve alone tops out at **2.5×**. Five age milestones (7, 30, 90, 180, 365 days) supply the last 0.5× at +0.1× each, **prorated linearly between milestones** — so no withdrawal ever falls off a cliff:

Lot age	Curve	Milestone bumps	Heat
0 days	1.00×	0	1.00×
7 days	1.21×	+0.10	1.31×
30 days	1.43×	+0.20	1.63×
90 days	1.75×	+0.30	2.05×
135 days	1.91×	+0.35	2.26×
180 days	2.05×	+0.40	2.45×
365 days	2.50×	+0.50	3.00×

The square-root shape is deliberate: early loyalty matters most, and diminishing returns keep a 2-year position from lapping a 1-year one into irrelevance.

The bumps belong to the lot's age, not to the badge. The Relic NFTs (section 9) are the soulbound record and carry the perks; the +0.1× bumps are computed from each lot's Heat age. That one rule settles three questions: a proportional withdrawal cools the bumps along with the age; a Deed buyer gets

the bumps their inherited age has earned and none of the seller's badges;
and
a reset wallet that still holds old Relics gets no bumps on new money until
that money ages.

Proportional cooling, defined per lot. Executing a withdrawal of fraction f of the position removes f of every lot's LP **and** multiplies every remaining lot's age by $(1 - f)$. Without the age step, a pro-rata draw would leave every lot's age untouched and the withdrawal would cool nothing.
Ages

are capped at 365 days **before** scaling, so a two-year lot that withdraws 50% lands at 182.5 days, exactly like a one-year lot; banked age past the curve can't absorb a withdrawal. Year two pays through Bedrock's perks (section 9), never through hidden age. Worked examples from a 365-day lot at 3.00x:

Withdraw	New age	Heat	Cost
1%	361.4 days	2.99x	-0.01x
10%	328.5 days	2.90x	-0.10x
50%	182.5 days	2.46x	-0.54x

Taps (section 9) are the exception: they draw the youngest lots first and cool nothing.

The global ceiling is $4.15x = 3.0 \text{ Heat} + 0.5 \text{ Capstone} + 0.15 \text{ Pyre} + 0.5$ conviction track — validated in `check:config` against the computed sum.
The conviction term is a cap on what the Conviction bucket can pay (section 10), so the ceiling holds even when few positions are sealed.

2. Ignition — the Genesis phase

The bootstrap. Default **120 hours** (configurable 48h–14d). **Nothing is ever locked — not even here:** principal can be withdrawn at any time during
Ignition, and the 3% genesis fee is not refunded

(withdrawDuringWindow: { enabled: true, refundFee: false }). The

genesis

fee applies in Ignition, in Kindling, and in every Fissure: **3% of every deposit goes straight to the treasury address — no splits, no router, no exceptions**. It is the treasury's only income before the Vent has fees to route. After close there is a 24-hour free exit window at 0% fee.

Pool	Asset	Weight
ign-usdc	USDC	100
ign-weth	WETH	100
ign-wbtc	WBTC	100
ign-eth-usdc-lp	WETH/USDC LP	150 (LP deliberately favored)

Rewards are the 41,000 ASH, streamed per second pro-rata on stake × time —

depositing early earns more. No vesting, no lockups on rewards. An **early-bird bonus** decays from **+10%** reward weight at the first second to **+0%** at close (linear) — urgency for the bootstrap without a first-block landgrab, on top of pro-rata rather than instead of it.

Ignition depositors receive two permanent things: a Heat head start and the

soulbound **First Flame** founder badge.

Where Ignition principal goes. EMBER doesn't exist until Kindling closes, and only USDC has a Forge pool among the four Ignition assets. So at close:

- USDC principal auto-stakes into **Cold Storage**, unless the depositor opted out at deposit.
- WETH, WBTC and WETH/USDC LP become claimable, fee-free.
- Every depositor's **Heat head start** (age dated to Ignition start) applies to the first Forge Deed they open or fund within 7 days of Kindling close, up to the USD value they kept in Ignition through close. A one-click "Pair and stake" zap on the claim screen turns WETH into Smelter LP

or USDC

into Blast Furnace LP, buying the EMBER half on Flow.

First Flame carries a **15-day phantom Heat floor** — even after a full withdrawal resets everything else, a founder's fresh stake accumulates from day 15 instead of day 0. Selling the wallet is the only way to sell the floor.

3. Kindling — the liquidity auction

Kindling replaces "the team seeds the pool" and "snipers own the open." It is a **24-hour clearing-price auction**: deposit USDC, everyone pays the same clearing price.

Parameter	Value
Duration	24 hours
Quote asset	USDC
EMBER seed	420,000 (one-time mint, 2% of max)
Raise range	\$100,000 – \$5,000,000
Per-wallet cap	\$25,000
Deposit fee	3% → treasury (the genesis fee)
Price mode	Clearing (single price for all)
LP destination	90% auto-staked Deeds owned by depositors · 10% burned forever
ASH bonus pool	1,000 ASH to participants (the Kindling leaf of the ASH tree)
Fallback	Treasury seeding (if min raise not hit; full refunds incl. fees)

The depositor accounting hole, closed. Burning *all* of the Kindling LP would leave depositors' USDC buying them nothing but a share of 1,000 ASH —

an unspecified donation. Instead: 90% of the Kindling LP becomes auto-staked

Deeds owned by the depositors (pro-rata to their raise contribution), and 10% is burned forever as the irrevocable liquidity floor. Not Bond LP, not governance-extractable, not a lock — the LP is theirs, staked, earning, and exitable under normal rules. This is the same mechanic Fissures use for satellite launches, so Kindling and Fissures are finally consistent.

What a Kindling depositor actually gets, stated before they deposit. For every \$100 in: \$3 goes to the treasury, the \$97 is paired with EMBER at the clearing price, and the depositor's Deeds hold 90% of that LP. That LP is two-sided: it holds **\$87.30 of USDC** plus an equal value of EMBER at the opening ratio — about **\$174.60 marked at the opening price**, plus their share of the 1,000 ASH. The pressure gauge shows both figures live and labels

them honestly: \$87.30 is the quote-side claim; \$174.60 is a mark, not a cash

exit — selling the EMBER half moves the price it was marked at, and minimum-

liquidity dilution, later fees and price changes are excluded. What the depositor gives up for an un-ruggable, un-sniped open is the 3% fee and the

10% of LP burned as the floor.

Atomic pool initialization: the deposit that pushes the timestamp past the deadline creates the pool, pairs the assets, mints the depositors' Deeds and

burns the 10% floor **in the same transaction**. Zero MEV window — no block exists in which the open can be sandwiched. The opening price mathematically

matches the auction's pressure gauge. The close is **permissionless**: anyone

can call it, and the caller earns a **0.05% bounty** from the Treasury's fee — bots race to close it the second it is due. No EOA to wait for, nothing to front-run.

4. The Forge — staking

The steady-state venue. Four pools:

Pool	Stakes	Emission weight	ASH weight
Blast Furnace	EMBER/USDC LP	1000	500
Smelter	EMBER/WETH LP	400	200
Ember Vault	single EMBER	400	300
Cold Storage	single USDC	200	0

LP pools carry the weight: the protocol rewards liquidity, not idle tokens. Cold Storage is for the risk-averse and earns no ASH — mercenary-stable capital gets base emissions and nothing else.

Fees: 1% to deposit, 1% to withdraw (taken on execution, never on arming), 0% to claim, **0% to compound — no timer, no age reset**. LP bought on the native Flow swap stakes with the deposit fee waived; waived zaps earn no Ashfall that epoch, so the waiver cannot be cycled for rewards.

Lots. Every deposit is its own lot with its own age (max 32 lots per position) — **per-lot accounting**. New money enters at 1.0× and never inherits old Heat; dust-aging is mathematically dead.

Provenance-preserving compounding. Rewards are tracked per lot, and when

Stoke executes, each lot's accrued rewards compound **back into that lot**. Rewards generated by lot A inherit lot A's age — never the blended position age, never 1.0×. A 365-day lot cannot launder a 2-day lot's rewards into 365-day Heat, and compounding never dilutes legitimately earned age. The only way to grow a hot position is to stoke it, and the age you stoke with is the age you earned.

Lot exhaustion. Stoke modifies its reward-producing source lots — it consumes no lot slot. Casting does the same (section 7): cast LP goes back into the lot that earned it, so it never needs a slot either. A genuine new

deposit past 32 lots triggers the UI's "Open New Deed" offer (Deeds are unlimited). Lots are **never** lossy-merged: averaging timestamps under the square-root Heat function would change reward entitlement.

Heat vs Streak are decoupled. Heat is the stake-weighted age that drives yield; Streak is the chronological clock (days since liquidity last left) that unlocks Relic perks. Adding money dilutes Heat but never touches the Streak. Withdrawing any amount resets the Streak to zero — but Heat only cools **proportionally**: withdraw 10%, keep 90% of your Heat age.

5. Deeds — time as an asset

Every position is a **Deed**, an NFT tradable on the in-dapp **Deed Market** for ETH or EMBER. A sold Deed keeps every token of liquidity in the pool — selling a position no longer means breaking LP and dumping. It means handing aged capital to a buyer.

On sale	What happens
LP, Capstone, conviction escrow + remaining term, unclaimed rewards	Transfer to the buyer
Heat age	Buyer inherits 80% — earned time stays worth more than bought time
Streak	Resets to 0 for the buyer
Badges (Relics, Pyre, First Flame)	Stay with the seller — reputation is earned by a person, age is a property of capital
Keystone	Dual : the seller's wallet keeps its soulbound Keystone Relic (the <i>person</i> completed the term); the Deed carries a permanent Seal Scar in its metadata (the <i>capital</i> completed one)
Fee	1% of the final price into the Vent (50% burn / 50% liquidity)

The sale is a 24-hour ascending auction. The seller sets a reserve price; the highest bid at close wins; below reserve, the Deed delists. No instant sales — and that is the point. Three exits, three prices:

Exit	Speed	Cost
Emergency exit	Instant, guaranteed	8%
Deed auction	24 hours, market-priced	1% + the market's discount
Withdrawal timer	72 hours (full exit)	1%

The auction answers the strongest objection to tradable positions — that an

instant 1% sale undercuts the priced escape hatch. It also kills fixed-price sniping: no bot can steal a mispriced listing. The market prices the exit instead of the protocol, but it cannot price it at zero time.

Anti-sniping: a valid bid inside the final 5 minutes extends the auction by 5 minutes, capped at 30 minutes total extension. Last-second sniping becomes last-minute bidding — the extension only triggers on genuine late competition.

The rules that keep it honest:

- Transfers outside the market are **refused by the token itself** — no OTC, no gifts, no dark pools. Every sale pays the Vent.
- A Deed with a pending Withdraw or Claim **cannot be listed**.
- Heat is **conserved-minus-haircut** on trade: it can move, never inflate. No sale creates a single point of Heat that wasn't earned by waiting.
- The Deed card prices honestly: intrinsic LP + Capstone value, impermanent loss against holding the two assets, and the Heat premium the market is paying — so bidders see exactly what the age costs.
- A wallet may hold any number of Deeds; deposits go into the Deed you choose. Laddering is native.

6. Capstone — the conviction sink

An optional EMBER-only lot inside each position. It earns no emissions and

never sells. It adds up to **+0.5× Heat** when the held EMBER is worth half your LP value, capped there so conviction can never outweigh a year of staying. Same 1% fee and same withdrawal timer to leave it. It touches neither Streak nor Heat age.

Stoking (compounding) fills the Capstone first, then mints LP whose quote side comes from the **match reserve** — fee revenue the Vent holds to pair with Stokes. A zap that sells half the reward is the fallback, and the UI says so before confirmation. **Stokes never sell by default.**

AutoStoke is opt-in per position — never open. The old design let anyone compound anyone's position for a 1% bounty; on a cheap-gas chain that pays bots to skim 1% of every position's yield continuously, front-running owners' own free stokes. Now, by default only the owner stokes. An owner who wants hands-off compounding enables keepers for that position with three limits: **tip** 0.1–1% (default 0.5%, contract-capped at 1%), **minimum interval** ≥24h (default 7d), **minimum size** (default 100 EMBER). A keeper stoke that violates any limit is rejected. `defaultOn: true` fails the build.

7. Casting — emissions that are liquidity

25% of every epoch's EMBER emission is never paid liquid. The protocol pairs it with quote asset from the match reserve at pool price, mints LP, and stakes it **back into the lot that earned it, at that lot's age**. That's the same provenance rule as Stoke: a cast is a Stoke the protocol does for you on a quarter of your rewards. The reward *is* liquidity: dumping it costs swap fees plus the 1% withdrawal fee plus the timer. A 2-day lot's cast lands at 2 days, so no reward can launder into old Heat — and because casting modifies

its source lots, it consumes no lot slot and needs no rolling lot with its lossy timestamp averaging.

Single-token pools. LP units cannot join an EMBER or USDC balance, so positions in the Ember Vault and Cold Storage receive their cast share as liquid EMBER. Casting into the source lot applies to LP pools only.

Ownership. The user's cast LP belongs to the source lot and exits through the normal timer. The match reserve's own contribution to the pairing is the protocol's, and that part is what burns as permanent liquidity (section 13).

Hard invariant: if the reserve cannot cover the quote side, the user receives liquid EMBER for the uncovered share. The protocol **never market-sells a user's emission** to complete a cast.

8. Pyre — burn-to-mint badges

Burn EMBER to the dead address; the Relics contract mints a soulbound badge:

Tier	Cumulative burn	Boost
Cinder	250 EMBER	+0.05× Heat
Wildfire	1,250 EMBER	+0.05× Heat
Conflagration	6,250 EMBER	+0.05× Heat

Permanent, account-wide, **capped at +0.15×** — below half the time ramp, so burning can never outweigh staying. The EMBER is gone, the badge cannot be sold: a pure deflationary sink that pays in loyalty instead of emissions. Registered Pyre burns count toward Ashfall at the 25% Pyre rate, so burners feed the pool they then earn from — without being able to rebate themselves.

9. Relics — the milestone NFTs

Soulbound, auto-minted at continuous-staking milestones. A Relic is the wallet's record and carries the perks below. The $+0.1\times$ Heat bump at each of the first five milestones is computed from lot age (section 1), so it cools, transfers and resets with the capital, while the badge stays with the person:

Relic	Days	Perks
Spark	7	Claim timer $\rightarrow$ 20h
Flame	30	Claim timer $\rightarrow$ 16h, Ashfall eligible
Blaze	90	Tap: withdraw 10% / 90d without resetting Streak
Inferno	180	Claim timer $\rightarrow$ 12h, Kindling priority
Eternal Flame	365	Tap 25% / 90d, 2 $\times$ governance weight, 25% Flow fee rebate
Bedrock	730	Tap 25% / 60d , 3 $\times$ governance weight, 50% Flow fee rebate

Taps draw the **youngest lots first**, so a Tap burns off your most dilutive deposits and *raises* blended Heat. On reset, badges remain as history but perks go dormant. Badges never transfer — and Deeds transfer *age*, never badges. **Bedrock** answers "what does year two pay?": the Heat ramp caps at 365 days, so the second year pays in perks, not multiplier.

10. Conviction tracks — sealed escrows

Opt-in bonus tracks layered **on top of** the click-timer, never replacing it. True **escrows**, not vesting balances:

Track	Length	Bonus
Ember	30 days	+0.15×
Forge	90 days	+0.25×
Kiln	180 days	+0.35×
Eternal	365 days	+0.50×

How the escrow works. Each epoch's total mint is **fixed before any position weights are evaluated** — then 10% of it is carved out as the **Conviction bucket**, and the remaining 90% splits 30/70 (27% of the epoch by stake, 63% by stake × Heat). Sealed positions compete for the bucket by stake × Heat × sealWeight; the ordinary share stays claimable as always, and the **bonus share accrues into escrow**.

The bucket is capped per position — additively. A seal bonus b is worth b extra multiplier units, so a sealed position's bucket payout can never exceed what b extra units would earn in the 70% pool that epoch: $\text{cap} = L \times \text{stake} \times b / W$, where L is the Heat pool and W is $\Sigma(\text{stake} \times \text{Heat})$. It is **not** $b \times$ the position's Heat-pool earnings — at Heat 3.65 that would pay 3.65× too much (1.825 units instead of 0.50). The reference oracle is `epochAllocation()` in the config. Whatever the cap holds back — and the whole bucket in an epoch when nobody is sealed — flows back into the Heat pool that same epoch. In an epoch with no seals the effective split is therefore **27% by stake / 73% by stake × Heat**, not 30/70. Without the cap, a single early sealer would collect the entire 10%, and the 4.15× ceiling would be a number in a doc. Conviction affects *distribution*, never *issuance*: seal bonuses cannot inflate the epoch. At term end the escrow pays out **into the Capstone by default** — the bonus never sells. Completing a term earns the **Keystone, dual**: a soulbound Keystone Relic for the wallet (proof the *person* completed the term — never transfers) and a permanent **Seal Scar** in the

Deed's metadata (proof the *capital* completed one — always travels with the Deed).

Breaking the seal. Withdrawing LP before term end forfeits the escrow — **never the principal**. The forfeited bonus routes **50% to the burn lane, 25% to the match reserve, 25% to the Sealed pot**, which streams to every position still inside its term. The faithless fund the faithful, and the principal still exits through the normal timer, unslashed.

11. Clocks — every timer in the protocol

Standing rule, honored everywhere: **timers start when the user clicks. Nothing is ever locked.**

Withdraw — progressive. Clicking "Start cooling" arms a visible countdown sized to the exit:

Exit size	Cooldown
≤ 10% of position	6 hours
≤ 25%	12 hours
≤ 50%	24 hours
≤ 75%	48 hours
100%	72 hours

At zero, the **green "Withdraw Now" button** — the only green in the app — goes live for a **24-hour execution window**. While cooling, **Heat and rewards are frozen**: the position neither ages nor earns. Cancel or miss the window and the position **resumes from the frozen Heat** — the cooling interval itself did not age, and no LP ever left, so no Heat is lost. Only actual *execution* triggers proportional cooling and Streak reset. The 1% fee is taken on execution. Each arm is per-lot.

Claim arms a 24-hour visible countdown (shortened by Relic perks), snapshots rewards at request time, then opens the green Claim Now. No

fee;
clocks untouched.

Compound is instant, free, and timeless — the only door with no lock.

Emergency exit skips every timer for **8%**: 50% burned, 25% to permanent liquidity, 25% to the stakers who stayed. The **deserter's tithe**: skipping the timer also forfeits the position's pending Ashfall into the rollover pool for the loyal. Both clocks reset.

12. Emissions, Ashfall, Quote yield

Each epoch's base emission splits **30/70**: 30% by raw stake (the newcomer floor), 70% by stake $\times$ Heat (the veteran gradient).

Ashfall re-emits prior-epoch burns (1-epoch lag) to stakers holding Flame or better, weighted by **stake $\times$ Heat²**. Recycle rates are **source-specific**: protocol-created burns (Forge fees, swap fees, emergency exits) count at **50%**; registered Pyre burns at **25%**; unsolicited dead-address transfers at **0%**. The 25% Pyre rate *reduces* self-recapture rather than eliminating it: a holder with eligible share p can still receive back $0.25 \times p$ of their own registered burn. The token disappears in every case; only the protocol's definition of "activity that funds stayers" changes. If nobody is eligible, Ashfall rolls forward — it never vanishes. Zero-fee Flow zaps are excluded. Ashfall is a mint, so it shares the epoch's headroom check with base emission: `totalMintForEpoch()` sizes base emission plus Ashfall together against `21M - totalSupply`, and both scale down pro-rata when headroom is short — Ashfall can never push supply past 21M on its own.

Quote yield: 5 bps of every Flow sell paid in the chain's **quote asset** (never EMBER) to the same Ashfall-eligible set at the same Heat² weight. Real yield beside emissions, not instead of them.

The Sealed pot: forfeited conviction escrows stream to positions still inside their term. Breaking your word pays everyone who kept theirs.

13. The Vent — the fee router

Every Forge fee (1% in, 1% out) splits **40% burn / 40% liquidity / 10% Hearth / 10% treasury** — must sum to 100, enforced in config validation.

The fee vault. Fees never hit the pool as they arrive. Every fee on a chain lands first in the chain's **fee vault**, part of the Vent. A permissionless `vent()` call processes the vault once it crosses **\$5,000** — and at least once per 24 hours regardless. Gas stays off user transactions, buys execute in batches, and no keeper is privileged.

Burn lane (`unwindAndBuyback`): fee LP is unwound, the EMBER half burned, the quote half buys back and burns — through an **1800-second TWAP** with **four-way buy protection**:

Guard	Value	Why
Average-price band	Spot within 2% of the 30-min average	A trader can't push the price and sandwich the burn inside one block
Slippage cap	Max 1% vs the average	Minimum-output discipline on every buy
Chunk size	No buy above 0.5% of the quote reserve	Whale buys wait for the next <code>vent()</code> call
Fail closed	Unsafe → the quote asset waits in the vault	Nothing is ever lost, only delayed

Allowlisted routers only; maximum 72-hour deferral. The build refuses a slippage cap above 1% or a TWAP window under 30 minutes.

Liquidity lane (`matchReserve`): fee LP is unwound, the EMBER half burned, the quote half held in the match reserve to pair with Stokes and Casting. Matched LP is burned as permanent liquidity. Idle reserve auto-pairs after

7
epochs.

Invariant: permanently burned liquidity must be full-range / non-repositionable. A burned concentrated-liquidity NFT can drift out of range into economically useless liquidity that nobody can ever rebalance. Concentrated venues may exist in Flow, but they are never the destination of the permanent-liquidity lane without an immutable, non-withdrawable rebalancer. The protocol AMM is full-range CPMM, so the invariant holds.

Token fees: Capstone in/out are paid in EMBER and burned at 100%.

Deed fees: 1% of every Deed auction's final price, 50/50 burn/liquidity.

14. Flow — the swap page

The native swap lists **EMBER, ASH, every configured single (WETH, USDC, WBTC...), and the LP tokens themselves** as first-class rows: buying the LP row zaps and mints, selling it unwinds, and staked LP cannot be touched until it exits through the timer. **Fees are asymmetric — leavers pay more than arrivers:**

Side	Total	LPs	Burn	Liquidity	Quote yield	Treasury
Buy EMBER	25 bps	20	2.5	2.5	—	0
Sell EMBER	35 bps	20	5	5	5	0

The 5 bps quote-yield slice exists **only on sells**: sellers explicitly fund stayer yield. The position card shows **impermanent loss next to Heat**. Eternal Flame holders receive a **25% rebate** on the protocol's fee share (50% at Bedrock), funded pro-rata from the burn/liquidity/quote-yield slices.

15. The Hearth

ASH staked in the Hearth earns the Vent's 10% fee share — real yield in fee assets, not emissions — plus governance weight (2× at Eternal Flame, 3× at Bedrock).

16. Treasury

Safe multisig (2-of; signers set at deploy), **48-hour timelock** on all parameter changes, on-chain reporting. The Safe deploys through its own deterministic factory (`0x4e1DCf7AD4e460CfD30791CCC4F9c8a4f820ec67`), recorded in config next to the salt nonce — so the treasury address is also decided before anything is deployed. Policy: **30%** POL seeding, **30%** match-reserve seeding (Ignition fees pre-fund day-one Stokes), **20%** incentives, **20%** buybacks. The Ignition 3% is the treasury's only income before fees exist. The Kindling close bounty (0.05%) comes from this fee.

17. Fissures — satellite launches

A satellite chain opens with a **Fissure**: a smaller Kindling seeded with EMBER the Treasury bridges over the OFT (250,000 default, counted against the cap), with the timelock approving the allocation. Same 3% fee, same 90/10 LP split, same atomic close. **Inferno+ holders on any chain get the first 24 hours to themselves**, then it opens to everyone — first access, never a head start: Heat, Streak and badges are chain-local and never bridge.

18. Referrals — disabled by default

A referee may attach a referrer's code; **3% of the referee's emission share is redirected** to the referrer while the referee holds Spark and the referrer holds Flame. Nothing is minted — **self-referral gains nothing net**. Both sides are strata-gated: recruiters must themselves be stayers. Off until the timelock decides the protocol wants recruiters.

19. Governance — two chambers

Launch runs on the timelock; two chambers take over later. The **Hearth Chamber** is ASH holders (the productive governance token); the **Forge Chamber** is liquidity × Heat (the stakers). Operational proposals use the appropriate chamber — Hearth matters to Hearth, staking matters to Forge.

Changes to **Treasury policy, new chains, Flow fee bands, emission curves,**
or

new contracts require both chambers. Neither Genesis ASH owners nor late

liquidity whales govern alone — and the two-token architecture finally has a

reason to exist beyond "ASH pays fees." Four things sit **above both chambers, out of reach of every vote, forever:** minting outside the Mantle, pausing withdrawals, touching user balances, redirecting the Vent.

20. Security

Contracts clamp fees at hard ceilings a rebrand can only lower, never raise:

5% deposits, 5% withdrawals, 5% genesis, 10% emergency,

100 bps swaps, 5% Deed sales. Deposits and swaps are pausable;

withdrawals are never pausable. Core contracts immutable; periphery upgradeable behind the timelock. Per-chain guardian addresses. **No audit has**

occurred — mainnet without a minimum of two independent audits is not on the table.

21. Omnichain

EMBER and ASH are LayerZero OFT v2: one canonical supply, native bridging, no

wrapped variants. Heat, Streak, Relics, and loyalty timers are **chain-local**

— bridging never carries Heat, which kills cross-chain multiplier gaming at the root.

Chain	Role	Emission share	LayerZero EID
Ethereum	home	45%	30101
BNB Chain	satellite	20%	30102
Base	satellite	15%	30184
Arbitrum One	satellite	10%	30110
Solana	satellite	10%	30168

The cross-chain gauge exists in config, disabled. Fixed emission shares run until it is proven. Fissures (section 17) are how satellites open.

22. Solana

Full Anchor program suite (staking program, fee-router program) with Metaplex soulbound Relics, sharing every fee, timer, and loyalty parameter from the same config. Program IDs and mint addresses are **ground keypairs** — the address is the public key, known before deploy, pinned with `declare_id!`. Vanity grinding for program prefixes is a planned step (solana-keygen).

23. Deterministic deployment

Every EVM contract deploys via **CreateX CREATE3** (`0xba5Ed099633D3B313e4D5F7bdc1305d3c28ba5Ed` — same on every chain). Salt = `deployer (20 bytes) ++ 0x00 ++ mined entropy (11 bytes) : the 0x00 flag` keeps **one identical address on every chain** while the deployer prefix means only that account can ever deploy it — un-front-runnable. The address depends only on factory + salt, never on bytecode or constructor args, so `pnpm mine:salts` runs **before any contract exists**. The miner is **idempotent**: re-running keeps every finalized entry and mines only what's missing.

Contract	Prefix	Address
EmberToken	F1E5	0xf1e5b7820a8c6f6349a576db516142c8d341dcc4
AshToken	A5E5	0xa5e55c164036f63e9b6e6337c465eedead49fb5b
FeeRouter	FEE5	0xfee5356949802d661959360a5e4004e1d8c685a2
Forge	F095	0xf09551c13e2ca9cbeed634452a1f37d89a25532d
Ignition	1917	0x19172ad5c06ccc86a6e2712d06500f1028cf185b
Kindling	5A1D	0x5a1d6ee608bf6e94b53789824705427ea40f2746
Hearth	EA27	0xea272470c2529afe408d13bf6057b72a1a8763ce
Relics	2E1C	0x2e1c923b32c214c4b7b2b8bd30daeb9f0a183dbd
Deeds	DEED	0xdeed54948eab9875d89a77cc383714d6a5d7dc6d

`contractAddress(name)` is the **only** way any module reads an address, and

it throws if a salt isn't mined. The deploy script recomputes every address from its salt and **aborts on mismatch**. Nothing writes to the config at deploy time, ever. `check:config` refuses any vanity prefix that isn't pure hex — and **refuses to build if any contract entry lacks a mined salt or address**: an unmined registry entry is a build error, not a runtime surprise.

`pnpm config:hash` prints keccak256 of the canonical config JSON. v5.3.1 changes config values, so the v5.3.0 hash is stale:

`0x9ecedb63d5d10951caaf506032a1cf6d14b9e2196f236e2d2a45c200870463f7`

(27,108 characters / 27,158 UTF-8 bytes; recursive key sort, array order preserved, compact `JSON.stringify`, Keccak-256 over UTF-8). The v5.3.0 hash `0x1554...94ed5` was 26,829 characters / 26,875 bytes — the earlier spec called the character count 'bytes'. Record it on-chain at deploy so every surface —

frontend, indexer, docs, deploy scripts — can prove it reads the same config

the contracts were initialized from.

Solana IDs (ground keypairs, keys/ — gitignored, back them up):

Program / Mint	Address
Staking program	HVQT6MycoarDw9ktbZBv282HH8SttSR4ycvTeSTw9kKe
FeeRouter program	ECgH9KpNFw9GoM9fe6uVYrmaJpQMsacA8EiiSKwo3Egu
EMBER mint	AtbnZK5DQibjpxkhynqq7617iuseoWaABytXN7Q3RnbD
ASH mint	7a2HrKABUqNEsPntziBaDMVMboQ6hThWJVFu9Zp3cAnW
Relics collection	GgpFhkNfS4Y16gX746NNzWJup6yDcpCRHjw5NNVKFdCL

Current mined addresses are bound to the placeholder deployer

`0x1111111111111111111111111111111111111111111111111111111111111111`. Two values finalize them: the real **DEPLOYER** EOA and the Treasury Safe **saltNonce**.

24. The config-first rule

Every brand string, fee, timer, pool, chain, page, piece of copy, and address lives in `furnace.config.ts`. Frontend, contracts, indexer, and docs all read

it; `pnpm check:config` fails the build on structural violations; `contracts` clamp anything the config tries to raise past a ceiling. A full rebrand — `names, tokens, theme, chains, pages` — is a one-file edit via `namePresets: names: namePresets.tephra` swaps the entire vocabulary.

The validator is **genuinely hostile by design**, and it is hostile in two different ways.

Exact, because the protocol's promises depend on them: the 3% genesis fee (Ignition, Kindling, Fissures), 1% deposit and 1% withdrawal fees; 0% compound; ASH allocations summing to exactly 70,000 with Ignition and Kindling matching their tree leaves; `supplyCeiling.hardCap =`

`token.maxSupply`

with Ashfall inside the same headroom budget; every split summing to 100;

per-lot accounting on; casting into the earning lot; the conviction bucket cap on; AutoStoke and referrals default-off; withdrawals never pausable; nothing locked (Ignition withdrawal open); the Vent 40/40/10/10; and complete

deploy identity (contract registry, real deployer, Safe nonce and signers, guardians, DVNs, Deeds salt).

Exact, because the protocol's identity depends on them: emission start 21,600/day, 365-day half-life, 1,000/day floor; 25% Casting; the 30/70 split; the 10% conviction bucket; the cooldown ladder; the 24-hour execution window; the 1800s TWAP floor and 1% slippage ceiling. These numbers *are* FURNACE —

a config that changes them is a new protocol, and the Rebrand Guide says so

explicitly. The validator does not bless forks as FURNACE.

"Single source of truth" is a build property, not a slogan.

25. Deliberate rejections

- **Transfer taxes** — break OFT bridging, aggregators, CEX listings; honeypot optics. Fees sit on actions, never on transfers.
- **Fixed lock durations** — the timer starts on click, always. Conviction escrows sit on top; they never replace.
- **Locking Ignition principal** — a transparent lock is still a lock. "Nothing is ever locked" holds in the bootstrap too.
- **33% early-exit slash** — a trap with better vocabulary. 8% prices the fast exit and pays the stakers who stayed; escrows forfeit only the bonus.
- **Dropping Deeds** — the "from first principles" simplification reasserts the old rejection without engaging the mitigations: 80% haircut, Streak reset, soulbound badges, Heat conserved-minus-haircut, and now the 24h

auction. Exits without sell pressure remain strictly better for stayers than exits with sell pressure.

- **Bond-LP tranches** — governance-redeemable liquidity is governance-extractable liquidity. Burned LP is un-ruggable.
- **Revenue-only emissions** — kept as the quote-yield slice, not the schedule. Starving loyal stakers in quiet months inverts the thesis.
- **Cross-chain loyalty sync** — chain-local accounting kills the gaming.
- **Killing ASH / the Hearth** — the two-token model is established and the Hearth gives ASH real fee yield.
- **Killing single-asset pools** — Cold Storage is a deliberate low-risk on-ramp that earns no ASH; its weight already says what we think of it.
- **Killing the Flow fee waiver** — documented, Ashfall-guarded, keeps liquidity formation inside the dapp.
- **Streak-based relic bumps** — harsher than proportional; contradicts "a 10% exit is never priced like a 100% exit."
- **Step-function milestone bumps** — the 365-day cliff priced a 1% exit at $-0.11\times$. Prorated bumps keep every exit smooth.
- **Rolling Cast lot** — timestamp averaging under sqrt Heat is the lossy merge the 32-lot rule bans. Casting into the earning lot kills it.
- **Uncapped conviction bucket** — one early sealer collecting the whole 10% breaks the $4.15\times$ ceiling. The per-position cap binds it.
- **Ashfall outside the headroom check** — a second mint outside the ceiling math reintroduces the breach v5.2 closed.
- **Range-validated economics** — exactness is the config-as-protocol guarantee. A rebrand that changes emissions or the Vent split is a new protocol; the validator doesn't bless it as FURNACE.
- **Flat 0.30% swap fee** — the 25/35 asymmetry is built, validated, and "sellers fund stayers" is cleaner attribution.
- **No hard cap** — the 21M cap costs nothing and "not infinite" is load-bearing optics.
- **Invalid vanity** (`0xFIRE...` , `1AVA` , non-hex prefixes, non-base58 Solana IDs) — the config refuses them at build time.

26. What's new in v5.3 (delta from v5.2.0)

Seven fixes and two completions. Every v5.2 mechanic stands; these make its

numbers agree with each other and with the config.

1. **Prorated milestone bumps.** The five $+0.1\times$ bumps now fill in linearly between milestones instead of stepping. The 365-day cliff (a 1% exit costing $-0.11\times$) is gone: $3.00\times \rightarrow 2.99\times$ on a 1% exit, $\rightarrow 2.90\times$ on 10%.
2. **Proportional cooling, defined per lot.** Withdrawing f removes f of every lot's LP and scales every lot's age by $(1 - f)$, capped at 365 days first. New `cooledAgeAfterWithdrawal()` helper; the worked examples are tested.
3. **Casting follows provenance.** Cast LP goes back into the lot that earned it, at that lot's age. The rolling Cast lot — and its lossy timestamp averaging — is gone.
4. **Conviction bucket capped per position.** A sealed position's bucket payout $\leq L \times \text{stake} \times \text{sealBonus} / W$ (additive — see section 10); the excess and any unsealed epoch flow back into the Heat pool. The $4.15\times$ ceiling binds.
5. **Ignition principal unlocked.** Withdrawable during Ignition (fee not refunded) — "nothing is ever locked" holds in the bootstrap. At close: USDC auto-stakes into Cold Storage (opt-out), other assets claimable fee-free, Heat head start rides on the first Deed opened within 7 days.
6. **One genesis fee.** The 3% now applies to Ignition, Kindling *and* Fissures — Kindling was missing it. Straight to treasury, no splits.
7. **Ashfall inside the headroom check.** `totalMintForEpoch()` sizes base emission + Ashfall against `21M - totalSupply` as one budget, pro-rata.
8. **Kindling depositor disclosure.** Per \$100: \$87.30 quote-side claim, ~\$174.60 two-sided mark at open — both shown live and labelled before deposit.
9. **namePresets** . `names: namePresets.tephra` swaps the entire vocabulary in one line — the Rebrand Guide's "one file, thirty minutes" is now real.

27. Before mainnet

1. Set the **DEPLOYER** EOA and Treasury Safe **saltNonce** → `pnpm mine:salts` finalizes all 9 addresses in seconds.
2. Set Treasury Safe **signers** (3-of-5 recommended) and per-chain **guardians**; configure ≥ 2 verified LayerZero **DVNs** per chain.
3. Professional audit – fee router + OFT integration + Deed Market + fee vault are the highest-risk surfaces. Minimum two independent audits.
4. Solana program audit, separately.
5. **Legal review of ASH and the Hearth**. A token that pays holders a share of protocol fees can be treated as a security in several jurisdictions. Get an opinion before ASH is sold, listed or farmed by the public.
6. Set `treasury.addresses` from the deployed Safe / Squads vault (the deploy script must assert equality), then record the final config hash on-chain.
7. Testnet run: 24-hour mock Ignition and Kindling; every partial withdrawal, cancellation, Stoke, multi-lot edge case, conviction expiry.
8. Invariant tests: fee sums, Ashfall roll-forward, matured-request unblockability, cap enforcement, prorated-bump continuity.
9. Grind Solana vanity program prefixes (optional, cosmetic).

FURNACE v5.3.1 · describes `furnace.config.ts` v5.3.1 · 2026-10-04 · hash

`0x9ecedb63d5d10951caaf506032a1cf6d14b9e2196f236e2d2a45c200870463f7`